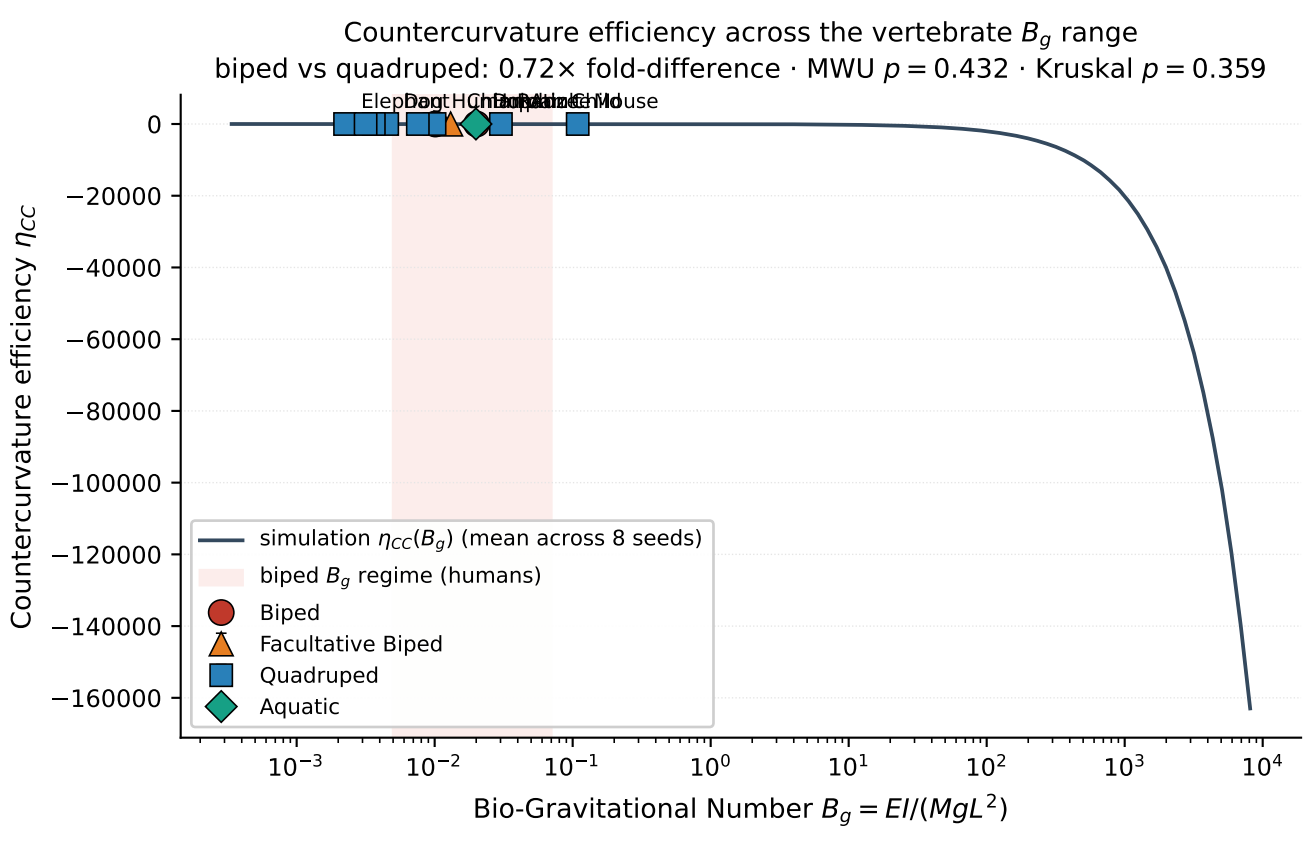
Countercurvature efficiency across the vertebrate B_g range
biped vs quadruped: 0.72× fold-difference · MWU $p = 0.432$ · Kruskal $p = 0.359$
Elephant
Dog
Human
Chimpanzee
Rabbit
Mouse
0
−20000
−40000
−60000
−80000
−100000
−120000
−140000
−160000
Countercurvature efficiency η_{CC}
simulation $\eta_{CC}(B_g)$ (mean across 8 seeds)
biped B_g regime (humans)
Biped
Facultative Biped
Quadruped
Aquatic
10^{-3}
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}
10^{4}
Bio-Gravitational Number $B_g = EI/(MgL^2)$